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Sub ject : DSA

Unit 3, S11, SLO1 - Circular Queue - Introduction

1. A circular queue connects the ____ position back to the front.
Answer: last (rear)

2. Circular queue overcomes the problem of ____ in a linear queue.
Answer: memory wastage (false overflow)

3. In a circular queue, when the rear reaches the end, it wraps around to index ____.
Answer: 0 (zero)

4. A circular queue follows the ____ principle.
Answer: FIFO (First In, First Out)

5. The condition for a full circular queue is: (rear + 1) % size == ____.
Answer: front

6. The condition for an empty circular queue is: front == ____.
Answer: -1

7. When the first element is inserted, both front and rear point to the ____ index.
Answer: 0th (first)

8. Circular queues are commonly used in ____ scheduling.
Answer: CPU round-robin / memory buffer

9. The ____ function checks if a circular queue is full.
Answer: isFull()

10. The ____ function checks if a circular queue is empty.
Answer: isEmpty()

11. Circular queues help in utilizing memory more ____ than linear queues.
Answer: efficiently

12. In a circular queue, incrementing the rear is done using ____ operator.
Answer: modulo (%)

13. Circular queues can be implemented using arrays or ____.
Answer: linked lists

14. The front element of a circular queue is accessed using the ____ pointer.
Answer: front

15. In circular queues, rear and front move in a ____ manner.
Answer: circular (clockwise)